Formulae used in straight line:

1) The direction ratios of the straight line $a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2$ are

$b_1c_2-b_2c_1,\ c_1a_2-c_2a_1,\ a_1b_2-a_2b_1$

2) If θ be the angle between the straight line

$\frac{x-x_1}{a_1}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$ and the plane $ax+by+cz+d=0$

then

$$\sin\theta=\frac{\Sigma aa_1}{\sqrt{\Sigma a^2}\sqrt{\Sigma a_1^2}}$$

& $$\cos\theta=\frac{\sqrt{\Sigma(ab_1-ba_1)^2}}{\sqrt{\Sigma a^2}.\sqrt{\Sigma a_1^2}}$$

3) The equation of the plane passing through the straight line $ax+by+cz+d=0=a'x+b'y+c'z+d'$ and parallel to the line $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ is given by

$$\frac{ax+by+cz+d}{al+bm+cn}=\frac{a'x+b'y+c'z+d'}{a'l+b'm+c'n}$$

শুভেচ্ছা বাণী

ঢাকা বিশ্ববিদ্যালয় গণিত অ্যালামনাই অ্যাসোসিয়েশন (DUMAA)-এর একাদশ বার্ষিক সাধারণ সভা-২০২২ ও দশম পুনর্মিলনী অনুষ্ঠিত হতে যাচ্ছে জেনে আমি আনন্দিত।

দেশের সর্বোচ্চ বিদ্যাপীঠ ঢাকা বিশ্ববিদ্যালয়ের গণিত বিভাগে অধ্যয়ন করেছেন তাঁদের মধ্যে নিবিড় যোগাযোগ থাকবে-এটাই সবার প্রত্যাশা। আজকের এই যান্ত্রিক সভ্যতা মানুষের আবেগকে সমাহিত করে শুধুই বেগকে বাড়িয়ে চলেছে। পারস্পরিক বিচ্ছিন্নতা ও আত্মকেন্দ্রিকতার দিকে ঠেলে দিচ্ছে। কিন্তু এ ধরনের পুনর্মিলনী অনুষ্ঠান ব্যক্তিকে সমষ্টির দিকে ধাবিত করে, আন্তরিকতায় উষ্ণতার সঞ্চার করে। দেশের বেশির ভাগ স্বনামধন্য গণিতবিদ এই বিভাগ থেকেই লেখাপড়া করেছেন। সেই সাথে অ্যাসোসিয়েশনের অনেক সদস্য সরকারি, বেসরকারি ও বিদেশী বিভিন্ন গুরুত্বপূর্ণ পদে আসীন হয়ে যোগ্যতা ও নিষ্ঠার সাথে দায়িত্ব পালন করছেন। সে কারণে তাঁদের যোগাযোগের মাধ্যম এই অ্যালামনাই অ্যাসোসিয়েশনটি অত্যন্ত গুরুত্ব বহন করে।

আমি অ্যাসোসিয়েশনের ১১শ বার্ষিক সম্মেলন ও ১০ম পুনর্মিলনী অনুষ্ঠানের সাফল্য কামনা করছি এবং সংশ্লিষ্ট সকলকে আন্তরিক অভিনন্দন ও শুভেচ্ছা জ্ঞাপন করছি।

জয়বাংলা। জয় বঙ্গবন্ধু ॥

ঢাকা বিশ্ববিদ্যালয় চিরজীবী হোক। বাংলাদেশ চিরজীবী হোক।

(অধ্যাপক ড. এ এস এম মাকসুদ কামাল)
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ঢাকা বিশ্ববিদ্যালয়

4) The condition of two straight lines

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \text{ and } \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$$

are coplanar is given by

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

If the two lines are coplaner, then the equ^{n} of the plane is given by

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

5) If the straight lines $ax+by+cz+d=0=a'x+b'y+c'z+d'$ and $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ are coplaner then

$$\frac{a\alpha+b\beta+c\gamma+d}{al+bm+cn} = \frac{a'\alpha+b'\beta+c'\gamma+d'}{a'l+b'm+c'n}.$$

6) If the straight lines $a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2$ and $a_3x+b_3y+c_3z+d_3=0=a_4x+b_4y+c_4z+d_4$ are coplanes then

$$\begin{vmatrix} a_1 & a_2 & a_3 & a_4 \\ b_1 & b_2 & b_3 & b_4 \\ c_1 & c_2 & c_3 & c_4 \\ d_1 & d_2 & d_3 & d_4 \end{vmatrix} = 0$$

7) The straight line $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ lies on the plane $ax+by+cz+d=0$ if

$$aa_1+bb_1+cc_1=0 \;\&\; ax_1+by_1+cz_1+d=0$$

8) The length of shortest distance between the straight lines $\frac{x-\alpha_1}{l_1} = \frac{y-\beta_1}{m_1} = \frac{z-\gamma_1}{n_1}$ and $\frac{x-\alpha_2}{l_2} = \frac{y-\beta_2}{m_2} = \frac{z-\gamma_2}{n_2}$ is given by

$$\frac{\begin{vmatrix} \alpha_2-\alpha_1 & \beta_2-\beta_1 & \gamma_2-\gamma_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}}{\sqrt{\sum (m_1n_2-m_2n_1)^2}}$$

(9) ~~If the three intersection lines of the planes~~ The condition that three planes $a_1x+b_1y+c_1z+d_1=0$, $a_2x+b_2y+c_2z+d_2=0$ and $a_3x+b_3y+c_3z+d_3=0$ ~~intersect at a point then~~ may have a common line of intersection if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

(10) Angle bet^n two SLs:

Let the eqn's of the given lines be

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \quad \text{---(1)}$$

& $$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \quad \text{---(2)}$$

The d.rs of the lines (1) & (2) are (a_1, b_1, c_1) & (a_2, b_2, c_2), respectively.

Let θ be the angle between them, then

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2} \cdot \sqrt{a_2^2+b_2^2+c_2^2}}$$

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Equation of a straight line :

Any equation of first degree in x, y, and z taken together always represents a straight line

General eqn of straight line :

Let us take two equations of first degree together

$$a_1x + b_1y + c_1z + d_1 = 0 \quad \&$$
$$a_2x + b_2y + c_2z + d_2 = 0$$

OR

$$\boxed{a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2} \quad \text{---(1)}$$

Any point which simultaneously satisfies both the equations given by (1), will lie on the curve of intersection of the planes (1)

Since two planes intersect in a straight line, therefore the equations (1) represent the equations of a straight line

The equations (1) are called the general equations of a straight line. It is generally written as

$$\left.\begin{aligned} a_1x + b_1y + c_1z + d_1 &= 0 \\ \& \; a_2x + b_2y + c_2z + d_2 &= 0 \end{aligned}\right\} \text{ or } (1)$$

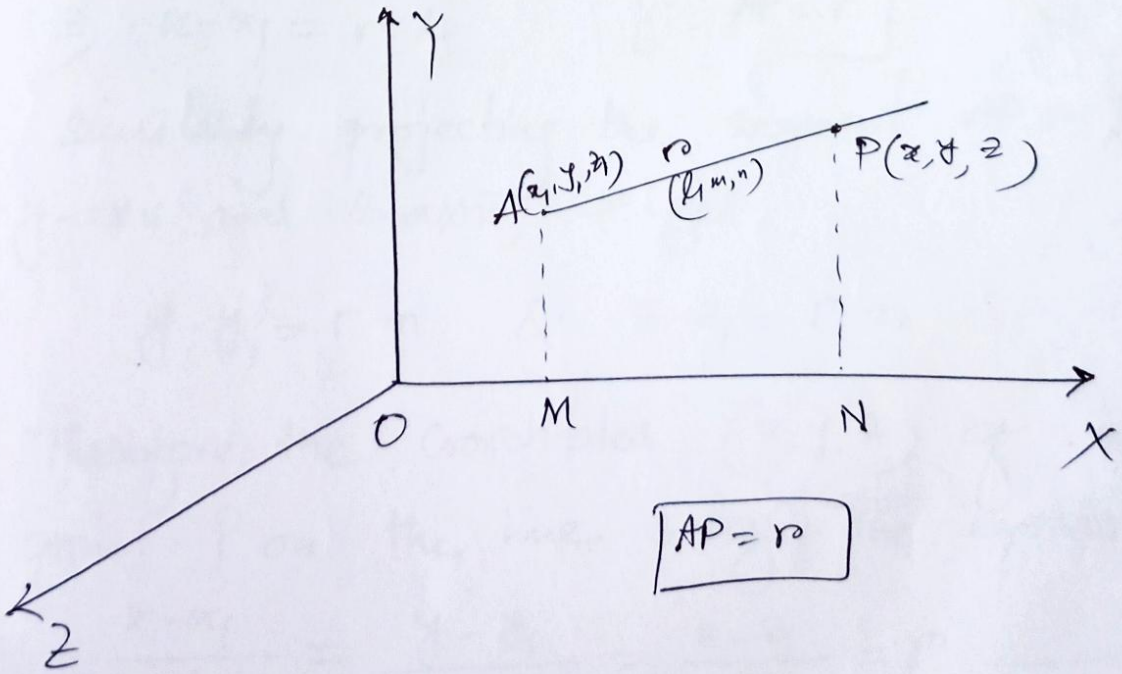
Symmetrical form of the equations of a straight line :

There are two types of symmetrical form of the equations of a straight line

(I) Symmetrical form in terms of dc^s &

(II) Symmetrical form in terms of dr^s

(I) To find the equations of a straight line which passes through a given point (x_1, y_1, z_1) and whose dc^s are (l, m, n) :



Let OX, OY, OZ be chosen rectangular coordinate axes. Let $A(x_1, y_1, z_1)$ be the given point on the line whose d.c.s are (l, m, n).

Choose a point $P(x, y, z)$ on the line at a distance r from the given point A.

Draw perpendiculars AM and PN on the x-axis from the points A and P, respectively.

So, MN is the projection of AP on the x-axis and is given by

$$MN = AP.\cos\alpha = AP.l$$

$$\Rightarrow x - x_1 = r.l \qquad [\because AP = r]$$

Similarly projecting the segment AP on the y-axis and z-axis, we get

$$y - y_1 = r.m \quad \& \quad z - z_1 = r.n$$

Therefore, the coordinates (x, y, z) of any point P on the line satisfy the equations

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r \qquad \text{---------- (1)}$$

Thus the equations (2) of the lines become

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} \quad \text{——— (3)}$$

The equations (3) are the required equations of the straight line

Equations of the coordinate axes Symmetrical form in terms of direction cosines:

The direction cosines of x-axis are 1, 0, 0

The coordinates of any point on the x-axis is given by $(x_1, 0, 0)$.

The equations of the straight line which passes through the point $(x_1, 0, 0)$ and whose d.c's are $(1, 0, 0)$ are given by

$$\frac{x-x_1}{1} = \frac{y-0}{0} = \frac{z-0}{0}$$

$\therefore$ Equations of x-axis

$$\boxed{\frac{x-x_1}{1} = \frac{y-0}{0} = \frac{z-0}{0}}$$

Similarly, equations of y-axis

$$\frac{x-0}{0} = \frac{y-y_1}{1} = \frac{z-0}{0}$$

And, equations of z-axis

$$\frac{x-0}{0} = \frac{y-0}{0} = \frac{z-z_1}{1}$$

Symmetrical form in terms of direction ratios :

Equations of x-axis

$$\frac{x-x_1}{a} = \frac{y-0}{0} = \frac{z-0}{0}$$

Equations of y-axis

$$\frac{x-0}{0} = \frac{y-y_1}{b} = \frac{z-0}{0}$$

Equations of z-axis

$$\frac{x-0}{0} = \frac{y-0}{0} = \frac{z-z_1}{c}$$

Standard equations of the three axes:

Equations of x-axis: $y = 0 = z$

Equations of y-axis: $x = 0 = z$

Equations of z-axis: $x = 0 = y$

Equations of a straight line through two given points:

Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be the given two points through which the line passes. The direction ratios of the line joining the two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ are

$$(x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1)$$

Hence, the required equations of the straight line are given by

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

OR

$$\frac{x - x_1}{x_1 - x_2} = \frac{y - y_1}{y_1 - y_2} = \frac{z - z_1}{z_1 - z_2}$$

Direction ratios of a straight line:

Let the equations of the straight line be

$$a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2 \quad \text{---(1)}$$

Then the equations of a line which is parallel to (1) and passes through the origin are

$$a_1x+b_1y+c_1z=0=a_2x+b_2y+c_2z \quad \text{---(2)}$$

Solving equations $a_1x+b_1y+c_1z=0$ and $a_2x+b_2y+c_2z=0$ we have

$$\frac{x}{b_1c_2-b_2c_1}=\frac{y}{c_1a_2-c_2a_1}=\frac{z}{a_1b_2-a_2b_1} \quad \text{---(3)}$$

which shows that the direction ratios of the line are

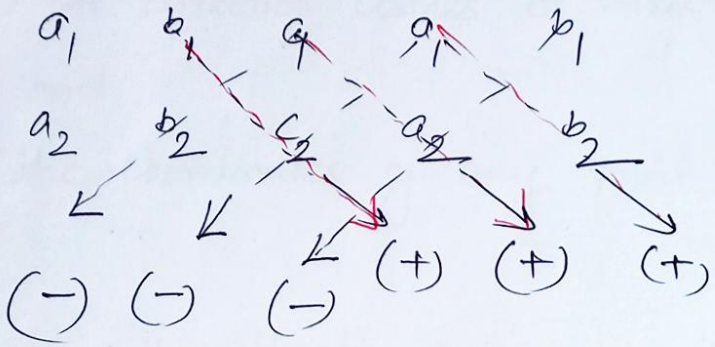
$$\boxed{(b_1c_2-b_2c_1,\ c_1a_2-c_2a_1,\ a_1b_2-a_2b_1)}$$

Short-cut method to find the direction ratios of a straight line:

Let the equations of the straight line be

$a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2$ ——(1)

Let (a,b,c) be the direction ratios of the line (1)



$$\left.\begin{aligned} a &= b_1c_2 - b_2c_1 \\ b &= c_1a_2 - c_2a_1 \\ c &= a_1b_2 - a_2b_1 \end{aligned}\right\}$$

To transform the general form of the equations of a straight line to symmetric form :

Let the general form of the equations of the straight line be given by

$$\left.\begin{aligned} a_1x+b_1y+c_1z+d_1 &= 0 \\ a_2x+b_2y+c_2z+d_2 &= 0 \end{aligned}\right\} \text{——(1)}$$

Now we are required to write down the symmetric form of the straight line given by eqn (1). For this we must need

(i) the direction cosines or direction ratios of the line and

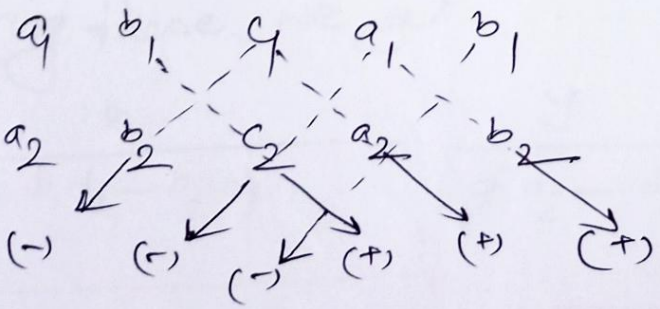
(ii) the coordinates of any point on the line

(i) To find the direction ratios of the line given by eqn (1) :

The equations of the straight line are

$$a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2 \quad \text{---(1)}$$

Let (a,b,c) be the dr's of the line (1)



$$\left.\begin{aligned} a &= b_1c_2-b_2c_1 \\ b &= c_1a_2-c_2a_1 \\ c &= a_1b_2-a_2b_1 \end{aligned}\right\}$$

(ii) To find the coordinates of a point on the line given by eqⁿ (1)

The coordinates of a point on a line can be chosen in many ways. One of these ways is that we choose the point as the one where the line cuts the xy-plane (i.e., $z=0$ plane), provided the line is not parallel to the xy-plane i.e., provided ~~that~~ $a_1b_2 - a_2b_1 \neq 0$

Putting $z=0$ in the eqⁿ given by (1), we have

$$\left.\begin{aligned} a_1x + b_1y + d_1 &= 0 \\ a_2x + b_2y + d_2 &= 0 \end{aligned}\right\} \quad \text{——— (2)}$$

Solving these, we get

$$\frac{x}{b_1d_2 - b_2d_1} = \frac{y}{d_1a_2 - d_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

$$\therefore x = \frac{b_1d_2 - b_2d_1}{a_1b_2 - a_2b_1} \quad \& \quad y = \frac{d_1a_2 - d_2a_1}{a_1b_2 - a_2b_1}$$

The line meets the plane $z=0$ at the point

$$\left(\frac{b_1 d_2 - b_2 d_1}{a_1 b_2 - a_2 b_1}, \frac{d_1 a_2 - d_2 a_1}{a_1 b_2 - a_2 b_1}, 0 \right)$$

Therefore, the equations of the given line in symmetric form are

$$\frac{x - \frac{b_1 d_2 - b_2 d_1}{a_1 b_2 - a_2 b_1}}{b_1 c_2 - b_2 c_1} = \frac{y - \frac{d_1 a_2 - d_2 a_1}{a_1 b_2 - a_2 b_1}}{c_1 a_2 - c_2 a_1} = \frac{z - 0}{a_1 b_2 - a_2 b_1}$$

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - 0}{c}$$

Note-1:

The coordinates of a point on a line can be chosen in many ways. One of these ways is that we choose the point as the one where the line cuts the yz-plane (i.e., $x=0$ plane) provided that the line is not parallel to the yz-plane i.e., $b_1 c_2 - b_2 c_1 \neq 0$

Note-2:

The coordinates of a point on a line can be chosen in many ways. One of these ways is that we choose the point as the one where the line cuts the zx-plane (ie, $y=0$ plane), provided that the line is not parallel to the zx-plane, ie; $a_1a_2 - c_2a_1 \neq 0$.

Note-3:

~~The coordinates of a point on a line can be chosen~~

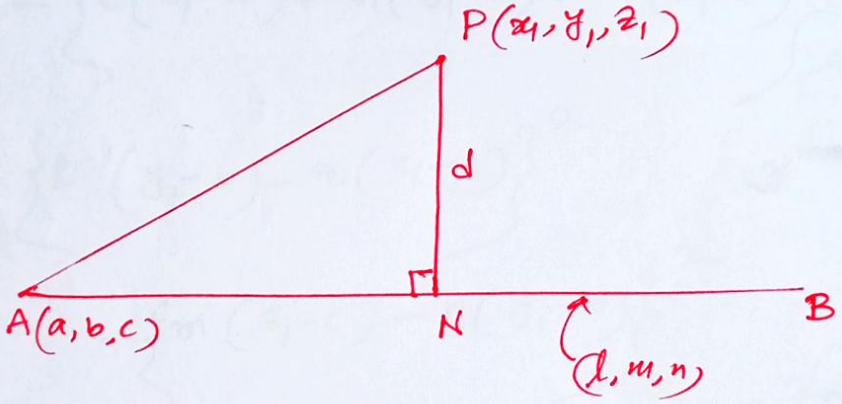
To find the coordinates of any two points on the line, we can write the eqn of the straight line in symmetric form.

Note-4:

By eliminating method, we can write the eqn of the straight line in symmetric form.

Distance of a point from a line:

Let d the length of a perpendicular distance from a point $P(x_1, y_1, z_1)$ to the line AB whose equations are $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$ with dc's (l, m, n) and passes through the point $A(a, b, c)$



Drow PN from P perpendicular to AB. Then we have

$$PN^2 = AP^2 - AN^2 \quad \text{---(1)}$$

But, $$AP^2 = (x_1-a)^2 + (y_1-b)^2 + (z_1-c)^2$$

And AN = Projection of AP on the given line AB

$$= l(x_1-a) + m(y_1-b) + n(z_1-c)$$

Thus,

$$PN^2 = \{(x_1-a)^2 + (y_1-b)^2 + (z_1-c)^2\} - \{l(x_1-a) + m(y_1-b) + n(z_1-c)\}^2$$

$$= \{(x_1-a)^2 + (y_1-b)^2 + (z_1-c)^2\}(l^2+m^2+n^2) - \{l(x_1-a) + m(y_1-b) + n(z_1-c)\}^2$$

$$= \{l(y_1-b) - m(x_1-a)\}^2 + \{m(z_1-c) - n(y_1-b)\}^2 + \{n(x_1-a) - l(z_1-c)\}^2$$

$$\Rightarrow d^2 = \begin{vmatrix} x_1-a & y_1-b \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1-b & z_1-c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1-c & x_1-a \\ n & l \end{vmatrix}^2$$

To find the coordinates of the foot of the perpendicular N :

Since the foot of the perpendicular N is a point on the line AB, the coordinates may be written as

$$(lr+a,\ mr+b,\ nr+c) \quad \text{——} \quad (*)$$

The d.r.s of the line PN are

$$(lr+a-x_1,\ mr+b-y_1,\ nr+c-z_1)$$

Again PN is perpendicular to AB

$$\therefore l(lr+a-x_1)+m(mr+b-y_1)+n(nr+c-z_1)=0$$

$$\Rightarrow r(l^2+m^2+n^2) = l(x_1-a)+m(y_1-b)+n(z_1-c)$$

$$\therefore r = l(x_1-a)+m(y_1-b)+n(z_1-c)$$

Putting the value of r into (*) we obtain the coordinates of N

———o———

Angle between two straight lines :

Let the equ^n's of the given lines be

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \quad \text{---------- (1)}$$

& $$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2} \quad \text{---------- (2)}$$

The dr^s of the lines (1) & (2) are (a_1, b_1, c_1) and (a_2, b_2, c_2), respectively.

Let θ be the angle between the two lines (1) and (2), then we have

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2} \cdot \sqrt{a_2^2+b_2^2+c_2^2}}$$

Condition of perpendicularity :

Two lines (1) & (2) will be perpendicular if $\theta = 90^\circ$, ie., $\cos\theta = \cos 90^\circ = 0$

$\therefore$ $\boxed{a_1a_2 + b_1b_2 + c_1c_2 = 0}$

Condition of parallelism :

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Corollary :

▷ The equations of the straight line, passing through the point (x_1, y_1, z_1) and perpendicular to x-axis are given by

$$\frac{x-x_1}{0} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

or $$x = x_1, \quad \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

▷ The equations of the straight line, passing through the point (x_1, y_1, z_1) and perpendicular to y-axis are given by

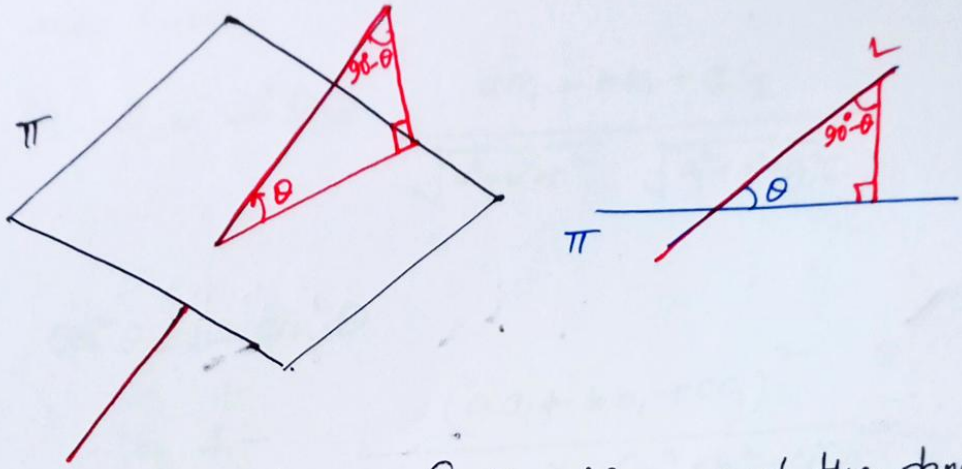
$$\frac{x-x_1}{a} = \frac{y-y_1}{0} = \frac{z-z_1}{c}$$

or $$y = y_1, \quad \frac{x-x_1}{a} = \frac{z-z_1}{c}$$

▷ The equations of the straight line, passing through the point (x_1, y_1, z_1) and perpendicular to z-axis are given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{0}$$ or $$z = z_1, \quad \frac{x-x_1}{a} = \frac{y-y_1}{b}$$

Angle between the plane & the straight line:



The given equations of the line and the plane are

$$\frac{x-a_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1} \quad \text{——— (1)}$$

$$ax+by+cz+d=0 \quad \text{——— (2)}$$

Angle between a line and a plane is the complement of the angle between the line and the normal to the plane

Here, the dr's of the normal to the given plane and the given straight line are (a, b, c) and (a_1, b_1, c_1), respectively.

Let θ be the angle between the line and the plane. So the angle between the line and the normal to the plane is $(90^\circ - \theta)$.

Thus, we have

$$\cos(90^\circ - \theta) = \sin\theta = \frac{aa_1 + bb_1 + cc_1}{\sqrt{a^2+b^2+c^2}\cdot\sqrt{a_1^2+b_1^2+c_1^2}}$$

Now

$$\cos^2\theta = 1 - \sin^2\theta$$

$$= 1 - \frac{(aa_1 + bb_1 + cc_1)^2}{(a^2+b^2+c^2)(a_1^2+b_1^2+c_1^2)}$$

$$= \frac{(a^2+b^2+c^2)(a_1^2+b_1^2+c_1^2) - (aa_1+bb_1+cc_1)^2}{(a^2+b^2+c^2)\cdot(a_1^2+b_1^2+c_1^2)}$$

$$\Rightarrow \cos^2\theta = \frac{(ab_1 - a_1b)^2 + (bc_1 - b_1c)^2 + (ca_1 - c_1a)^2}{(a^2+b^2+c^2)\cdot(a_1^2+b_1^2+c_1^2)}$$

$$\therefore \cos\theta = \frac{\sqrt{(ab_1 - a_1b)^2 + (bc_1 - b_1c)^2 + (ca_1 - c_1a)^2}}{\sqrt{a^2+b^2+c^2}\cdot\sqrt{a_1^2+b_1^2+c_1^2}}$$

Condition of perpendicularity:

The straight line is perpendicular to the plane, it means that the line is parallel to the normal of the plane. Hence we have

$$\frac{a_1}{a} = \frac{b_1}{b} = \frac{c_1}{c}$$ ← Required Condition

Condition of parallelism:

The straight line ① and the plane ② will be parallel if $\theta = 0$ or $\sin\theta = \sin 0 = 0$

$$\therefore \quad aa_1 + bb_1 + cc_1 = 0$$

Note:

The straight line is parallel to the plane means that the line is perpendicular to the normal of the plane.

Condition that the line may lie in the plane:

The given equations of the line and the plane are

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} \quad \text{——— (1)}$$

& $ax+by+cz+d=0$ ——— (2)

From (1) we have

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r \text{ (say)}$$

$\therefore$ $x = x_1+lr$, $y = y_1+mr$, $z = z_1+nr$ } OR { $x = lr+x_1$, $y = mr+y_1$, $z = nr+z_1$ }

Therefore, the coordinates of any point of the line is

$$(lr+x_1,\ mr+y_1,\ nr+z_1).$$

If the line lies in the plane, then for all values of r, the point $(lr+x_1, mr+y_1, nr+z_1)$ should lie on the given plane $ax+by+cz+d=0$.

Thus we have

$$a(lr+x_1)+b(mr+y_1)+c(nr+z_1)+d=0$$

$$\Rightarrow r(al+bm+cn)+(ax_1+by_1+cz_1+d)=0 \quad \text{——— (3)}$$

Straight Line Related Problems

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Problem:

Find the equation of the ~~plane~~ line through the point $(2,1,3)$ and perpendicular to the plane $3x+4y-z=7$.

Sol^n:

We know the equ^n the line passing through the point (x_1, y_1, z_1) and having dr^s (a,b,c) is given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

Hence, the equ^n of the SL that passes through the point $(2,1,3)$ is given by

$$\frac{x-2}{a} = \frac{y-1}{b} = \frac{z-3}{c} \quad \text{——— (1)}$$

Since line (1) is perpendicular to the given plane $3x+4y-z=7$, so the line is parallel to the normal of that plane.

Therefore,

$$\frac{a}{3} = \frac{b}{4} = \frac{c}{-1} = k \text{ (say)}$$

$\Rightarrow a = 3k,\ b = 4k,\ c = -k$

Now, substituting the values of a, b, c into (1) we obtain

$$\frac{x-2}{3k} = \frac{y-1}{4k} = \frac{z-3}{-k}$$

$$\Rightarrow \frac{x-2}{3} = \frac{y-1}{4} = \frac{z-3}{-1}$$

which is the required equⁿ of the ~~plane~~ line.

Problem:

Find the equation of the line through the point $(1,2,3)$ and is parallel to the two planes

$$x-y+2z-5=0 \quad \& \quad 3x+y+z-6=0.$$

Solⁿ:

We know that the equⁿ of the SL passing through the point (x_1, y_1, z_1) and having drˢ (a,b,c) is given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} \quad \text{——— (1)}$$

Given, the equⁿs of the planes are

$$x-2y+2z-5=0 \quad \text{——— (2)}$$

$$\& \quad 3x+y+z-6=0 \quad \text{——— (3)}$$

Since line ① is parallel to the planes ② & ③, so line ① is perpendicular to the normals of the planes ② & ③, respectively.

Hence,

$$a - b + 2c = 0 \quad \text{——— ④}$$

$$\& \quad 3a + b + c = 0 \quad \text{——— ⑤}$$

Solving ④ & ⑤ using cross-multiplication we get

$$\frac{a}{-1-2} = \frac{b}{6-1} = \frac{c}{1+3} = k \text{ (say)}$$

$$\Rightarrow \frac{a}{-3} = \frac{b}{5} = \frac{c}{4} = k$$

$$\therefore a = -3k, \; b = 5k, \; c = 4k$$

Substituting the values of a, b, c into ① we get

$$\frac{x-1}{-3k} = \frac{y-2}{5k} = \frac{z-3}{4k}$$

$$\Rightarrow \frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$$

which is the required eqn of the SL.

Problem:

Find the equⁿ of the plane through the straight line $3x-4y+5z-10=0=2x+2y-3z-4$ and parallel to $x=2y=3z$.

Solⁿ:

Given,

$$3x-4y+5z-10=0=2x+2y-3z-4 \quad \text{——— (1)}$$

&

$$x=2y=3z$$

$$\Rightarrow \frac{x}{6}=\frac{y}{3}=\frac{z}{2} \quad \text{——— (2)}$$

Let the equⁿ of the plane through the line (1) is given by

$$3x-4y+5z-10+k(2x+2y-3z-4)=0 \quad \text{— (3a)}$$

$$\Rightarrow (3+2k)x+(2k-4)y+(5-3k)z-10-4k=0 \quad \text{— (3b)}$$

The plane (3b) is parallel to the SL (2), hence the SL is perpendicular to the normal of the plane.

$$\therefore 6(3+2k)+3(2k-4)+2(5-3k)=0$$

$$\Rightarrow 18+12k+6k-12+10-6k=0$$

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$\Rightarrow 12k + 16 = 0$

$\Rightarrow k = -\frac{4}{3}$

Substituting the value of k into (3a) to have

$$3x - 4y + 5z - 10 - \frac{4}{3}(2x + 2y - 3z - 4) = 0$$

$$\Rightarrow 9x - 12y + 15z - 30 - 8x - 8y - 12z + 16 = 0$$

$$\Rightarrow x - 20y + 27z - 14 = 0$$

Ans)

Problem:

Determine the equⁿ of the straight line through the origin in the plane $x - y - z = 0$ and perpendicular to the straight line $\frac{x}{2} = \frac{y}{1} = \frac{z}{-3}$.

Solⁿ:

We know that the equⁿ of the SL passing through the origin and having drˢ (a, b, c) is given by

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c} \quad \text{—(1)}$$

Since equⁿ (1) lies on the plane $x - y - z = 0$ and is perpendicular to the SL $\frac{x}{2} = \frac{y}{1} = \frac{z}{-3}$, so

we have

$$a - b - c = 0 \quad \text{——— (2)}$$

& $$2a + b - 3c = 0 \quad \text{——— (3)}$$

Solving (2) & (3) using cross-multiplication we get

$$\frac{a}{3+1} = \frac{b}{-2+3} = \frac{c}{1+2} = k \text{ (say)}$$

$$\Rightarrow \frac{a}{4} = \frac{b}{1} = \frac{c}{3} = k$$

$$\therefore a = 4k,\ b = k,\ c = 3k$$

Substituting the values of a, b, c into (1) we get

$$\frac{x}{4k} = \frac{y}{k} = \frac{z}{3k}$$

$$\Rightarrow \frac{x}{4} = \frac{y}{1} = \frac{z}{3}$$

which is the required eqn of the SL.

—o—

Problem:

Find the equⁿ of the plane through the line $3x-4y+5z-10=0=2x+2y-3z-4$ and is parallel to the line $x=2y=3z$.

Solⁿ:

Given,

$$3x-4y+5z-10=0=2x+2y-3z-4 \quad \text{——— (1)}$$

& $x=2y=3z$

$$\Rightarrow \frac{x}{6}=\frac{y}{3}=\frac{z}{2} \quad \text{——— (2)}$$

Let the equⁿ of any plane through the line (1) is given by

$$3x-4y+5z-10+k(2x+2y-3z-4)=0$$

$$\Rightarrow (3+2k)x+(2k-4)y+(5-3k)z+(-10-4k)=0 \quad \text{——— (3)}$$

If the plane (3) is parallel to the line (2), then

$$6(3+2k)+3(2k-4)+2(5-3k)=0$$

$$\Rightarrow 18+12k+6k-12+10-6k=0$$

$$\Rightarrow 12k=-16 \quad \Rightarrow k=\frac{-4}{3}$$

Substituting the value of k into (3) we get

$$3x-4y+5z-10-\frac{4}{3}(2x+2y-3z-4)=0$$

$$\Rightarrow x-20y+27z-14=0$$

which is the required eqⁿ of the plane.

Problem:

Find the magnitude and the eqⁿ of the S.D between the lines $\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$ and $\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1}$. Also find the coordinates of the points where the S.D meets the given lines.

Solⁿ:

Given the eqⁿs of SLs are

$$\frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1} \quad \text{---------- (1)}$$

&

$$\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1} \quad \text{---------- (2)}$$

From ① we can write

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} = \lambda_1 \text{ (say)}$$

$\therefore x = \lambda_1 + 3$

$y = -2\lambda_1 + 5$

$z = \lambda_1 + 7$

Any point on line ① is given by

$$P(\lambda_1 + 3, -2\lambda_1 + 5, \lambda_1 + 7)$$

Similarly, from ② we get

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} = \lambda_2 \text{ (say)}$$

$\therefore x = 7\lambda_2 - 1$

$y = -6\lambda_2 - 1$

$z = \lambda_2 - 1$

Any point on line ② is given by

$$Q(7\lambda_2 - 1, -6\lambda_2 - 1, \lambda_2 - 1)$$

Let the S.D line intersects the line ① at the point $P(\lambda_1 + 3, -2\lambda_1 + 5, \lambda_1 + 7)$ and line ②

at the point $Q(7\lambda_2-1, -6\lambda_2-1, \lambda_2-1)$.

Therefore, the dr's of the S.D line are given by

$$(a,b,c) \equiv (\lambda_1+3-7\lambda_2+1, -2\lambda_1+5+6\lambda_2+1, \lambda_1+7-\lambda_2+1)$$

$$\Rightarrow (a,b,c) \equiv (\lambda_1-7\lambda_2+4, -2\lambda_1+6\lambda_2+6, \lambda_1-\lambda_2+8)$$

Since, the S.D. line is perpendicular to the lines ① & ②, simultaneously

$$\therefore (\lambda_1-7\lambda_2+4)\cdot 1 + (-2\lambda_1+6\lambda_2+6)(-2) + (\lambda_1-\lambda_2+8)(1) = 0$$

$$\Rightarrow \lambda_1-7\lambda_2+4+4\lambda_1-12\lambda_2-12+\lambda_1-\lambda_2+8=0$$

$$\Rightarrow 6\lambda_1-20\lambda_2=0$$

$$\Rightarrow \lambda_1 = \frac{10}{3}\lambda_2 \quad \text{—— ③}$$

Also,

$$7(\lambda_1-7\lambda_2+4)-6(-2\lambda_1+6\lambda_2+6)+1(\lambda_1-\lambda_2+8)=0$$

$$\Rightarrow 7\lambda_1-49\lambda_2+28+12\lambda_1-36\lambda_2-36+\lambda_1-\lambda_2+8=0$$

$$\Rightarrow 20\lambda_1-86\lambda_2=0$$

$$\Rightarrow \lambda_1 = \frac{43}{10}\lambda_2 \quad \text{— ④}$$

Solving ③ & ④ we get $\lambda_1 = 0$ & $\lambda_2 = 0$

$\therefore$ P(3,5,7) & Q(-1,-1,-1) are the two points where the S.D line intersect the lines

$\therefore$ Magnitude/length of the S.D line

$$= \sqrt{(3+1)^2 + (5+1)^2 + (7+1)^2}$$

$$= \sqrt{116}$$

$$= 2\sqrt{29}$$

Ans)

Equ^n of the S.D line is given by

$$\frac{x-3}{3+1} = \frac{y-5}{5+1} = \frac{z-7}{7+1}$$

$$\Rightarrow \frac{x-3}{4} = \frac{y-5}{6} = \frac{z-7}{8}$$

$$\Rightarrow \frac{x-2}{2} = \frac{y-5}{3} = \frac{z-7}{4}$$

Problem:

Find the S.D between lines $\frac{x+3}{2} = \frac{y+5}{3} = \frac{z+7}{-3}$ and $\frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1}$. Comment on whether the given SLs are coplanar or not.

Sol^n:

Given SLs are:

$$\frac{x+3}{2} = \frac{y+5}{3} = \frac{z+7}{-3} \quad \text{---} (1)$$

& $$\frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1} \quad \text{---} (2)$$

Let (a, b, c) are the d^{r}s of the S.D line. Since the S.D line is perpendicular to both the lines (1) & (2), so we have

$$2a + 3b - 3c = 0 \quad \text{---} (3)$$

& $$4a + 5b - c = 0 \quad \text{---} (4)$$

Solving (3) & (4) using the cross-multiplication technique, we get

$$\frac{a}{-3+15} = \frac{b}{-12+2} = \frac{c}{10-12}$$

$$\Rightarrow \frac{a}{12} = \frac{b}{-10} = \frac{c}{-2}$$

$$\Rightarrow \frac{a}{6} = \frac{b}{-5} = \frac{c}{-1} = k \text{ (say)}$$

$$\therefore a = 6k, \quad b = -5k, \quad c = -k$$

Let (l, m, n) be the dc's of the S.D line, then

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{6k}{\sqrt{(6k)^2+(-5k)^2+(-k)^2}}$$

$$= \frac{6k}{k\sqrt{36+25+1}}$$

$$\therefore l = \frac{6}{\sqrt{62}}$$

Similarly,

$$m = \frac{-5}{\sqrt{62}}, \quad n = \frac{-1}{\sqrt{62}}$$

Any arbitrary point on line ① is $P(-3, -5, -7)$

Any " " " " ② is $Q(-1, -1, -1)$

∴ Length of the S.D line = Projection of the line joining the two points P & Q on the S.D. line

$$= \left| l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1) \right|$$

$$= \left| \frac{6}{\sqrt{62}}(-1+3) - \frac{5}{\sqrt{62}}(-1+5) - \frac{1}{\sqrt{62}}(-1+7) \right|$$

$$= \frac{1}{\sqrt{62}} \left| 12 - 20 - 6 \right|$$

$$= \frac{14}{\sqrt{62}}$$

(Ans)

Since the S.D. between the lines ① & ② is not zero and the lines are not parallel, so the lines ① & ② are not coplanar.

Problem:

Find the magnitude and the eqⁿ of the S.D line between the lines whose equations are

$\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-36}{-6}$ and $x+y=0, z=4$.

Solⁿ:

Given the eqⁿs of SLs are

$$\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-36}{-6} \quad \text{——— ①}$$

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& $x+y=0,\ z=4$ ——— (2)

Let us rewrite equⁿ (2) as follows

$$\left.\begin{aligned} x+y+0(z-4) &= 0 \\ 0.x+0.y+1.(z-4) &= 0 \end{aligned}\right\}$$

Hence, the symmetric form of the equⁿ (2) is now given by

$$\frac{x}{1} = \frac{y}{-1} = \frac{z-4}{0} \quad \text{——— (3)}$$

Let (a,b,c) be the drs of the S.D line

Since the S.D line is perpendicular to both the lines (1) & (3), so we have

$$4a+3b-6c=0 \quad \text{——— (4)}$$

& $a-b=0$ ——— (5)

From (4) & (5) using cross-multiplication we get

$$\frac{a}{0-6} = \frac{b}{-6} = \frac{c}{-7}$$

$$\Rightarrow \frac{a}{6} = \frac{b}{6} = \frac{c}{7} = k \text{ (say)}$$

$$\therefore a = 6k,\ b = 6k,\ c = 7k$$

Let (l,m,n) be the dc's of the S.D line, then

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{6k}{\sqrt{(6k)^2+(6k)^2+(7k)^2}}$$

$$= \frac{6k}{\sqrt{36k^2+36k^2+49k^2}}$$

$$= \frac{6}{11}$$

Similarly, $m = \frac{6}{11}$ & $n = \frac{7}{11}$

Any arbitrary point on line ① is $P(1,2,36)$

" " " " " ③ " $Q(0,0,4)$

∴ Length of the S.D line = Projection of the line joining the two points P & Q on S.D. line

$$= \left| l(x_2-x_1) + m(y_2-y_1) + n(z_2-z_1) \right|$$

$$= \left| \frac{6}{11}(1-0) + \frac{6}{11}(2-0) + \frac{7}{11}(36-4) \right|$$

$$= \left| \frac{6+12+224}{11} \right| = 22 \text{ Units}$$

Ans

Now, the eqn of the S.D line is given by

$$\begin{vmatrix} x-1 & y-2 & z-36 \\ 4 & 3 & -6 \\ 6 & 6 & 7 \end{vmatrix} = 0 = \begin{vmatrix} x & y & z-4 \\ 1 & -1 & 0 \\ 6 & 6 & 7 \end{vmatrix}$$

$$\Rightarrow 57(x-1) - 64(y-2) + 6(z-36) = 0$$

$$\Rightarrow 57x - 64y + 6z - 145 = 0$$

&

$$-7x - 7y + 12(z-4) = 0$$

$$\Rightarrow 7x + 7y - 12z + 48 = 0$$

$$\therefore 57x - 64y + 6z - 145 = 0 = 7x + 7y - 12z + 48$$

which is the required eqn's of the S.D line.

Problem:

Find the length and equⁿ of the S.D line between the two given lines $x+y+2z+5=0=x+2y+2z+4$ and $x+2y+z+2=0=x+y+3z+5$.

Solⁿ:

Given,

$$x+y+2z+5=0=x+2y+2z+4 \quad \text{——(1)}$$

$$x+2y+z+2=0=x+y+3z+5 \quad \text{——(2)}$$

Let the equⁿ of the plane through the lines (1) is given by

$$x+y+2z+5+k_1(x+2y+2z+4)=0$$

$$\Rightarrow (1+k_1)x+(1+2k_1)y+(2+2k_1)z+4k_1+5=0 \quad \text{——(3)}$$

Similarly, the equⁿ of the plane through the lines (2) is given by

$$x+2y+z+2+k_2(x+y+3z+5)=0$$

$$\Rightarrow (1+k_2)x+(2+k_2)y+(1+3k_2)z+5k_2+2=0 \quad \text{——(4)}$$

If the planes (3) & (4) are parallel, then

$$\frac{1+k_1}{1+k_2} = \frac{1+2k_1}{2+k_2} = \frac{2+2k_1}{1+3k_2} = \lambda \text{ (say)}$$

$\therefore 1+k_1 = \lambda(1+k_2) \Rightarrow 1+k_1-\lambda-\lambda k_2 = 0 \quad -(5)$

$1+2k_1 = \lambda(2+k_2) \Rightarrow 1+2k_1-2\lambda-\lambda k_2 = 0 \quad -(6)$

$2+2k_1 = \lambda(1+3k_2) \Rightarrow 2+2k_1-\lambda-3\lambda k_2 = 0 \quad -(7)$

(6) − 2 * (5)

$-1+\lambda k_2 = 0 \quad -(8)$

(7) − 2 * (5)

$\lambda - k_2\lambda = 0$

$\Rightarrow k_2\lambda = \lambda$

$\Rightarrow \boxed{k_2 = 1} \quad [\because \lambda \neq 0]$

(8) $\Rightarrow \boxed{\lambda = 1}$

Also, $\boxed{k_1 = 1}$

Substituting the values of k_1, k_2 & λ into (3) & (4) we get respectively

$$2x + 3y + 4z + 9 = 0 \quad \text{——} (9)$$

& $$2x + 3y + 4z + 7 = 0 \quad \text{——} (10)$$

The eqn's (9) and (10), i.e., the eqn's of the planes respectively pass through the lines represented by (1) & (2).

Since the planes (9) & (10) are parallel to each other so the distance between them itself is the S.distance. (S.D)

$\therefore$ Length of the S.D line $= \left| \dfrac{d_1 - d_2}{\sqrt{a^2+b^2+c^2}} \right|$

$$= \left| \frac{9-7}{\sqrt{2^2+3^2+4^2}} \right|$$

$$= \frac{2}{\sqrt{4+9+16}}$$

$$= \frac{2}{\sqrt{29}}$$

Ans)

If (3) is perpendicular to either (9) or (10), then

$$2(1+k_1)+3(1+2k_1)+4(2+2k_1)=0$$

$$\Rightarrow 2+2k_1+3+6k_1+8+8k_1=0$$

$$\Rightarrow 16k_1+13=0$$

$$\Rightarrow k_1=-\frac{13}{16}$$

Substituting the value of k_1 into (3) we obtain

$$3x-10y+6z+28=0$$

Again if (4) is perpendicular to either (9) or (10) then we have

$$2(1+k_2)+3(2+k_2)+4(1+3k_2)=0$$

$$\Rightarrow 2+2k_2+6+3k_2+4+12k_2=0$$

$$\Rightarrow 17k_2=-12$$

$$\Rightarrow k_2=-\frac{12}{17}$$

Substituting the value of k_2 into (4) we obtain

$$5x+22y-19z-26=0$$

Hence, the eqn's of the S.D line is given by

$$3x-10y+6z+28=0=5x+22y-19z-26$$

Ans

Problem:

Show that the length of the S.D between the lines $\frac{x+3}{-2} = \frac{y-6}{3} = \frac{z-3}{2}$ and $\frac{x}{2} = \frac{y-6}{-2} = \frac{z}{1}$ is 3 and that the line of S.D cuts z-axis.

Sol^n:

Given the equ's of the SLs are

$$\frac{x+3}{-2} = \frac{y-6}{3} = \frac{z-3}{2} = k_1 \text{ (say)} \quad \text{---(1)}$$

$$\frac{x}{2} = \frac{y-6}{-2} = \frac{z}{-1} = k_2 \text{ (say)} \quad \text{---(2)}$$

Any arbitrary point on the line (1) is given by

$$P(-2k_1-3,\ 3k_1+6,\ 2k_1+3)$$

Any arbitrary point on the line (2) is given by

$$Q(2k_2,\ -2k_2+6,\ -k_2)$$

Let (a,b,c) be the d.r^s of the S.D line, then

$$(a,b,c) \equiv (2k_2+2k_1+3,\ -2k_2+6-3k_1-6,\ -k_2-2k_1-3)$$

$$\equiv (2k_2+2k_1+3,\ -2k_2-3k_1,\ -k_2-2k_1-3)$$

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Since, the S.D line is perpendicular to both the lines ① & ②, then

$$-2(2k_2+2k_1+3)+3(-2k_2-3k_1)+2(-k_2-2k_1-3)=0$$

$$\Rightarrow -4k_2-4k_1-6-6k_2-9k_1-2k_2-4k_1-6=0$$

$$\Rightarrow -17k_1-12k_2-12=0$$

$$\Rightarrow 17k_1+12k_2+12=0 \quad \text{——} ③$$

Similarly,

$$2(2k_2+2k_1+3)-2(-2k_2-3k_1)-1(-k_2-2k_1-3)=0$$

$$\Rightarrow 4k_1+3k_2+3=0 \quad \text{——} ④$$

Solving ③ & ④ we obtain

$$k_1=0 \ \& \ k_2=-1$$

Therefore, $P(-3,6,3)$ & $Q(-2,8,1)$ are the points where the S.D line meet at two respective planes.

∴ S.D = Distance between the two points $P(-3,6,3)$ & $Q(-2,8,1)$

$$\Rightarrow S.D = \sqrt{(-2+3)^2 + (8-6)^2 + (1-3)^2}$$

$$= \sqrt{1+4+4}$$

$$= 3$$

[Showed]

Now, eqn of the S.D line be

$$\frac{x+3}{-1} = \frac{y-6}{-2} = \frac{z-3}{2} = k \text{ (say)}$$

$\therefore$ Any arbitrary point on the S.D line be

$$M(-k-3, -2k+6, 2k+3)$$

Again let the eqn of the z-axis be

$$\frac{x}{0} = \frac{y}{0} = \frac{z}{1} = \lambda \text{ (say)}$$

$\therefore$ Any arbitrary point on z-axis be

$$N(0,0,\lambda)$$

If S.D line & z-axis intersect with each other then

$$-k-3 = 0 \Rightarrow k = -3$$
$$-2k+6 = 0 \Rightarrow k = -3$$
$$2k+3 = \lambda \Rightarrow \lambda = -3$$

Hence, the coordinates of the point on the z-axis be $(0, 0, -3)$

Thus, S.D line intersects z-axis.

[Showed]

Problem:

Show that the eqn of the plane passing through the points $(2, -1, 0)$ and $(3, -4, 5)$ and parallel to the straight line $2x = 3y = 4z$ is given by

$$29x - 27y - 22z - 85 = 0.$$

Sol^n:

Let the eqn of the plane passing through the point $(2, -1, 0)$ is given by

$$a(x-2) + b(y+1) + c(z-0) = 0 \quad \text{---------- (1)}$$

Also, the eqn of the straight line is

$$2x = 3y = 4z$$

$$\Rightarrow \frac{x}{6} = \frac{y}{4} = \frac{z}{3} \quad \text{---------- (2)}$$

The plane (1) is parallel to the SL (2), therefore the normals to the plane (1) is perpendicular to (2)

$\therefore\ 6a+4b+3c=0$ ——— (3)

Again, plane (1) passes through the point $(3,-4,5)$, therefore

$$a(3-2)+b(-4+1)+c(5-0)=0$$

$$\Rightarrow a-3b+5c=0 \quad \text{——— (4)}$$

Eliminating a, b, c from equ^{n}s (1), (3), (4) gives

$$\begin{vmatrix} x-2 & y+1 & z \\ 6 & 4 & 3 \\ 1 & -3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(20+9)+(y+1)(3-30)+z(-18-4)=0$$

$$\Rightarrow 29(x-2)-27(y+1)-22z=0$$

$$\Rightarrow 29x-27y-22z-58-27=0$$

$$\Rightarrow 29x-27y-22z-85=0$$

[Showed]

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Problem:

Prove that the equation to the plane containing the line $\frac{y}{b}+\frac{z}{c}=1,\ x=0$ and parallel to the line $\frac{x}{a}-\frac{z}{c}=1,\ y=0$ is $\frac{x}{a}-\frac{y}{b}-\frac{z}{c}+1=0$ and if 2d is the shortest distance (S.D) then also prove that

$$\frac{1}{d^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}.$$

Sol^n:

Let the equation of the plane containing the SL $\frac{y}{b}+\frac{z}{c}=1,\ x=0$ is the given by

$$\left(\frac{y}{b}+\frac{z}{c}-1\right)+kx=0 \qquad \text{——— (1a)}$$

$$\Rightarrow kx+\frac{1}{b}y+\frac{1}{c}z-1=0 \qquad \text{——— (1b)}$$

Given,

$$\frac{y}{b}+\frac{z}{c}=1,\quad x=0 \qquad \text{——— (1)}$$

&

$$\frac{x}{a}-\frac{z}{c}=1,\quad y=0 \qquad \text{——— (2)}$$

The symmetric form of the equ^n (2) is given by

$$\frac{x}{a}=\frac{y}{0}=\frac{z+c}{c} \qquad \text{— (3)}$$

Since (3) is parallel to (1b), then we have

$$ak + 0 \cdot \frac{1}{b} + c \cdot \frac{1}{c} = 0$$

$$\Rightarrow ak + 1 = 0$$

$$\Rightarrow k = -1/a$$

Substituting the value of k into (1b) to have

$$-\frac{1}{a}x + \frac{1}{b}y + \frac{1}{c}z - 1 = 0$$

$$\Rightarrow \frac{1}{a}x - \frac{1}{b}y - \frac{1}{c}z = -1$$

$$\Rightarrow \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0 \quad \text{——— (4)}$$

[Proved]

From (3) we can write,

$$\frac{x}{a} = \frac{y}{0} = \frac{z+c}{c} = \lambda \text{ (say)}$$

$$\therefore x = a\lambda, \quad y = 0, \quad z = c\lambda - c$$

For $\lambda = 1$, $(a, 0, 0)$ be any point on the line (3)

Let $2d$ be the S.D from any arbitrary point

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$(a,0,0)$ to the plane (4), then we have

$$2d = \left| \frac{\frac{a}{a} + 0 + 0 + 1}{\sqrt{\left(\frac{1}{a}\right)^2 + \left(-\frac{1}{b}\right)^2 + \left(\frac{1}{c}\right)^2}} \right|$$

$$\Rightarrow 2d = \frac{2}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\Rightarrow 4d^2 = \frac{4}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$\Rightarrow d^2 = \frac{1}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$\Rightarrow \frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

[Proved]